

FL→GL commanded by the node-local Disaster-FD, with the agent's watchdog (API 1.7.0)

ReSIDS agent

Disaster-FD
(node-local monitor)

Disaster-FD
(federated, regional)

two clocks from Disaster-FD: $round = local\ tick$ · $fed_round = federated\ tick$ (here every 5 local ticks)

local tick r : inactivity observed by the node-local Disaster-FD
local evidence only — no regional quorum

alt

[node-local Disaster-FD alive]

set_mode {to: gossip, reason: node_failure} (local IPC)

result {applied, effective at the next round boundary}

commit FL→GL @ $r + detect_lag$ (local ticks)
(measured: round 102, with or without a slower federated tick)

architecture_change {FL→GL} decided_by: monitor

event federated to the regional view

alt

[node-local Disaster-FD down]

✗ local FD process crashed / hung

watchdog: no command from its own FD for D local ticks

commit FL→GL @ $r + detect_lag + D$
(measured: round 105, $D = 3$)

architecture_change {FL→GL, reason: autonomous_fallback} decided_by: autonomous

the ONLY autonomous action — towards the safe mode only; the event is read on the next since= poll

later

[GL→FL — federated Disaster-FD decides]

membership full for dwell FEDERATED ticks
(regional evidence)

set_mode {to: federated, reason: recovery} (fan-out to all 15 · unanimity · on a fed_round tick)

commit GL→FL on a federated tick
(measured: 606 with every tick federated → 630 with a tick every 5)

result {applied}

no fallback of any kind: if the federation cannot decide, the system rests in GL